REGRESSION ASSIGNMENT

1) Identify your problem statement

This problem goal is to build to predict the model that estimates the **insurance charges(output or target)**. It based on multiple input such as **Age, Sex, BMI, No. of Children and smoking status** columns in dataset.

- Input and output data is clearly mentioned in this dataset. So this is **supervised learning.**
- Check the output variable type. Here output variable column is **insurance charges**. This is continuous numerical column in the dataset. So this is **Regression.**

2) Identify your problem statement

From the provided dataset like:

- Total Rows - 1338
- Total Columns – 6

Column Details:

- Age - Numerical
- Sex - Categorical (male/Female)
- BMI – Numerical
- Children – Numerical
- Smoker – Categorical (Yes/No)
- Charges – Numerical (output variable)

3) Data preprocessing steps

The data preprocessing steps needed for categorical data like **Sex & Smoker** column. Here, convert nominal data into numerical data. This conversion can be done in **one-hot encoding**.

- Sex -> encode as male = 1 & female = 0.
- Smoker-> encode as yes = 1 & no = 0.

To find following the machine learning regression method using in r2 value

1. **MULTIPLE LINEAR REGRESSION** (R2 value)=0.78947
2. **SUPPORT VECTOR MACHINE**

S.NO	HYPER PARAMETER	LINEAR (r value)	RBF (Non-Linear) (r value)	POLY (r value)	SIGMOID (r value)
1	C=10	0.46246	-0.03227	0.03871	0.03930
2	C=100	0.62887	0.32003	0.61795	0.52761
3	C=500	0.76310	0.66429	0.82636	0.44460
4	C=1000	0.76493	0.81020	0.85664	0.28747
5	C=2000	0.74404	0.85477	0.86055	-0.59395
6	C=3000	0.74142	0.86633	0.85989	-2.12441

The **SVM Regression** R2 Value (RBF Non-linear and hyper parameter (c=3000)) = 0.86633

3. **DECISION TREE**

S.NO	CRITERION	MAX FEATURES	SPLITTER	R VALUE
1	squared_error	None	best	0.70097
2	squared_error	None	random	0.74580
3	squared_error	sqrt	best	0.72487
4	squared_error	sqrt	random	0.65262
5	squared_error	log2	best	0.74671
6	squared_error	log2	random	0.71293
7	friedman_mse	None	best	0.70290
8	friedman_mse	None	random	0.72672
9	friedman_mse	sqrt	best	0.71134
10	friedman_mse	sqrt	random	0.73215
11	friedman_mse	log2	best	0.63994
12	friedman_mse	log2	random	0.68363
13	absolute_error	None	best	0.65906
14	absolute_error	None	random	0.69151
15	absolute_error	sqrt	best	0.77188
16	absolute_error	sqrt	random	0.62675
17	absolute_error	log2	best	0.72647
18	absolute_error	log2	random	0.75135
19	poisson	None	best	0.71410
20	poisson	None	random	0.71408
21	poisson	sqrt	best	0.69197
22	poisson	sqrt	random	0.69116
23	poisson	log2	best	0.74541
24	poisson	log2	random	0.64437

The **DECISION TREE** R2 Value (absolute_error,sqrt,best) = 0.77188.

4. RANDOM FOREST

S.NO	CRITERION	MAX FEATURES	N_ESTIMATORS	R VALUE
1	squared_error	None	10	0.85390
2	squared_error	None	100	0.85284
3	squared_error	sqrt	10	0.85571
4	squared_error	sqrt	100	0.85020
5	squared_error	log2	10	0.85290
6	squared_error	log2	100	0.85733
7	friedman_mse	None	10	0.85334
8	friedman_mse	None	100	0.85364
9	friedman_mse	sqrt	10	0.85761
10	friedman_mse	sqrt	100	0.85434
11	friedman_mse	log2	10	0.85193
12	friedman_mse	log2	100	0.85414
13	absolute_error	None	10	0.85331
14	absolute_error	None	100	0.85382
15	absolute_error	sqrt	10	0.85269
16	absolute_error	sqrt	100	0.85425
17	absolute_error	log2	10	0.85678
18	absolute_error	log2	100	0.85493
19	poisson	None	10	0.85241
20	poisson	None	100	0.84940
21	poisson	sqrt	10	0.85401
22	poisson	sqrt	100	0.85382
23	poisson	log2	10	0.85371
24	poisson	log2	100	0.85311

The **RANDOM FOREST** R2 Value (friedman_mse,sqrt,10) = 0.85761.

RESULT:

The **SVM Regression** R2 Value (RBF Non-linear and hyper parameter (c=3000)) = 0.86633. This is the best regression model.